

One-form Sobolev linear-frame completion and terminal-frame kernel leakage

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-070 reduced the remaining regular finite-cutoff problem to a weighted linear production frame and a coupled nonlinear shifted residual. It proposed to put the raw non- pp symmetric coefficient in $H^{-1-\kappa}$ at $\kappa = 1/10$, pay it by R-068, and then use the terminal current square on the nonlinear remainder.

Both proposed steps require correction. The raw symmetric coefficient has a nonzero strict low-high principal term and is only in $H^{-3/2-\delta}$ after taking its spatial divergence. The same issue affects the R-070 regularity attribution of the pure- pp and Cartan currents. Covariance centering does not remove this nonzero-frequency term. This note proves the correct fixed-floor matrix one-form lift in $H^{-1/2-\delta}$, extends the R-068 payment to every $1 < s < 2$, and thereby closes the complete symmetric and Cartan linear frame with larger but finite probability moments.

The nonlinear terminal square also has a genuine frame kernel. An exact production fixture makes the remaining nonlinear coefficient term affine and unbounded along a derivative-displacement direction invisible to that square. Thus the next theorem must be an integrated Cameron–Martin/strict-past estimate for the kernel leakage, not a pointwise Schur absorption.

All statements use the hash-pinned A1 covariance on the fixed torus, the A7 covariance-normal convention, fixed $e > 0$, and common scalar real-even spectral approximations. The linear-frame completion is a continuum model statement. The kernel fixture is finite-dimensional and exact for every $e > 0$. No finite-energy one-use or Nelson conclusion is asserted.

2 Exact weighted one-form and its polynomial core

Use the R-070 notation

$$\rho = |U|^2, \quad m_r = U^T S_r U, \quad q_r = \frac{m_r}{\rho + e}, \quad p_r = 2S_r U, \quad v_r = 2(S_r - q_r I)U,$$

$$P_{r,i} = p_r \cdot \partial_i U, \quad V_{r,i} = v_r \cdot \partial_i U,$$

and

$$K_{r,p} = 2S_r, \quad K_{r,v} = 2(S_r - q_r I) - \frac{v_r U^T + U v_r^T}{\rho + e}.$$

The complete symmetric current preceding the R-070 divergence is

$$\begin{aligned} \mathcal{J}_{r,i}^{\text{lin}} &= q_{11} P_{r,i} K_{r,p} + q_{12} (P_{r,i} K_{r,v} + V_{r,i} K_{r,p}) \\ &\quad + q_{22} V_{r,i} K_{r,v}, \end{aligned} \tag{2.1}$$

and

$$\mathbb{Q}_r^{\text{lin}} = -\frac{1}{2} \sum_i \partial_i \mathcal{J}_{r,i}^{\text{lin}}. \tag{2.2}$$

Put $R_i = \partial_i \rho$ and $H_r = K_{r,v} - K_{r,p}$. Directly,

$$V_{r,i} = P_{r,i} - q_r R_i, \quad (2.3)$$

$$\begin{aligned} P_{r,i} K_{r,v} &= P_{r,i} K_{r,p} + P_{r,i} H_r, \\ V_{r,i} K_{r,p} &= P_{r,i} K_{r,p} - q_r R_i K_{r,p}, \\ V_{r,i} K_{r,v} &= P_{r,i} K_{r,p} + P_{r,i} H_r - q_r R_i K_{r,p} - q_r R_i H_r. \end{aligned} \quad (2.4)$$

Consequently the full polynomial core of (2.1), not merely its visibly labelled pp summand, is

$$(q_{11} + 2q_{12} + q_{22}) P_{r,i} K_{r,p}. \quad (2.5)$$

At the production point

$$2q_{12} + q_{22} = 0.00609374999999848 \neq 0, \quad q_{11} + 2q_{12} + q_{22} = 0.0105937499999974. \quad (2.6)$$

Thus the nominal non- pp block contains a nonzero copy of the same polynomial current. Equation (2.5) is useful bookkeeping, but extracting it does not improve the generic regularity of the remaining rational one-form.

Entrywise there is a smooth matrix one-form Θ_r such that

$$(\mathcal{J}_{r,i}^{\text{lin}})_{\mu\nu} = \Theta_{r,\mu\nu,a}(U) \partial_i U^a. \quad (2.7)$$

The positive floor makes all derivatives of Θ_r smooth with at most polynomial growth. This is the correct model class.

3 The raw one-tenth attribution is false

Take a nonzero constant low-frequency background x and a high-frequency Gaussian perturbation z_n . Near the origin at fixed floor,

$$q_r(x) = O(|x|^2/e), \quad v_r(x) = p_r(x) + O(|x|^3/e), \quad K_{r,v}(x) = K_{r,p} + O(|x|^2/e).$$

Hence the nominal non- pp one-form coefficient has the principal part

$$\Theta_r^{\text{np}}(x) = (2q_{12} + q_{22}) p_r(x) K_{r,p} + O(|x|^3/e), \quad (3.1)$$

which is nonzero for a suitable r, x by (2.6). The strict low-high contribution to (2.2) then contains

$$-\frac{1}{2} \Theta_r^{\text{np}}(x) \Delta z_n. \quad (3.2)$$

Since the production covariance is comparable to $\langle n \rangle^{-4}$, the Fourier variance of (3.2) is bounded below and above by nonzero constants on a fixed cone. Therefore

$$\mathbb{E} \|\mathbb{Q}_{r,\Lambda}^{\text{np}}\|_{H^{-1-\kappa}}^2 \gtrsim \sum_{|n| \leq \Lambda} \langle n \rangle^{-2-2\kappa} \asymp \Lambda^{1-2\kappa}. \quad (3.3)$$

The raw attribution fails for $\kappa \leq 1/2$; at $\kappa = 1/10$ the growth is $\Lambda^{4/5}$. Subtracting a same-point covariance changes only the zero-frequency contraction channel and cannot cancel (3.2).

The pure polynomial term has the same low-high principal symbol, so the R-070 statement that the raw pure- pp Laplacian is an $H^{-1-1/10}$ centered derivative-square model is also withdrawn. Likewise, the Gaussian coefficient in the Cartan term is a first-derivative one-form current with generic regularity $H^{-1/2-\delta}$, not $H^{-1/10}$.

R-063 does not contradict this conclusion. Its improved jets explicitly remove the low-value/high-root region and retain the omitted lower chaoses in the exact grouped reconstruction. They cannot be relabelled as the literal full product in (2.7).

4 Fixed-floor matrix one-form Sobolev lift

4.1 Statement

Let U be a centered stationary finite-dimensional Gaussian field on $\mathbb{T}_L^3$ with even matrix covariance

$$c\langle k\rangle^{-4}I \leq \widehat{C}(k) \leq C\langle k\rangle^{-4}I. \quad (4.1)$$

Let $U_\Lambda = R_\Lambda U$, where R_Λ is any admitted common real-even scalar approximation from R-050. Let $\Theta : \mathbb{R}^d \rightarrow \mathbb{R}^{m \times d}$ be smooth, with every derivative of polynomial growth. Define smooth currents

$$\mathcal{J}_{\Theta,i}^{(\Lambda)} = \Theta(U_\Lambda) \partial_i U_\Lambda. \quad (4.2)$$

Matrix one-form Sobolev lift. For every finite p and every $\delta > 0$, (4.2) converges to one regulator-independent canonical geometric current in

$$L^p(\Omega; H^{-1/2-\delta}). \quad (4.3)$$

The current is a continuous reconstruction from the R-050 enhanced field $(U, \mathbb{X})$, where

$$\mathbb{X}_i^{ab} = U^a \circ \partial_i U^b. \quad (4.4)$$

No first-derivative current counterterm occurs in the admitted regulator class.

4.2 Proof

Fix $\alpha \in (1/3, 1/2)$. R-050 gives, in every finite probability moment,

$$U \in \mathcal{C}^\alpha \cap H^{1/2-\epsilon}, \quad \partial_i U \in H^{-1/2-\epsilon}, \quad (4.5)$$

and the Fourier–Wick estimate

$$\mathbb{E}|\widehat{\mathbb{X}_i^{ab}}(n)|^2 \lesssim \langle n \rangle^{-3}. \quad (4.6)$$

Thus $\mathbb{X}_i^{ab} \in H^{-\epsilon}$ in every finite probability moment, and the coupled-cutoff tails converge there. Gaussian hypercontractivity upgrades the second-moment estimate to every finite p .

Bony decomposition first puts the strict low–high term in the target space:

$$T_{\Theta(U)} \partial_i U \in H^{-1/2-\epsilon}, \quad (4.7)$$

because $\Theta(U)$ is bounded pathwise and its bound has every finite moment. For the only nonclassical resonant term, first-order parilinearisation gives

$$\Theta_a(U) = T_{\partial_b \Theta_a(U)} U^b + R_{\Theta,a}, \quad R_{\Theta,a} \in \mathcal{C}^{2\alpha}. \quad (4.8)$$

The leading resonance is reconstructed as

$$\partial_b \Theta_a(U) \mathbb{X}_i^{ba} \quad (4.9)$$

plus the standard paraproduct commutator. Multiplication in (4.9) is classical after choosing $\epsilon < \alpha$. The remainder resonance has regularity $3\alpha - 1 > 0$. The high–low paraproduct is no rougher than (4.7). The same L^2 Littlewood–Paley estimates apply to matrix entries; only finitely many entries are present.

Equations (4.7)–(4.9) define a locally Lipschitz map from

$$(U, \mathbb{X}) \mapsto \mathcal{J}_\Theta \quad \text{in } H^{-1/2-\delta}. \quad (4.10)$$

Polynomial-growth coefficient bounds are integrable because the Gaussian $\mathcal{C}^\alpha$ norm has every finite moment. Convergence and scheme independence of the R-050 enhancement therefore imply (4.3). Finally, Gaussian integration by parts gives

$$\mathbb{E}[\Theta_a(U_\Lambda)\partial_i U_\Lambda^a] = \mathbb{E}[\partial_b \Theta_a(U_\Lambda)] \text{Cov}(U_\Lambda^b, \partial_i U_\Lambda^a) = 0, \quad (4.11)$$

because the common covariance is even. This proves the no-current- counterterm assertion.

The corresponding covariance-normal divergence identity is worth recording. At every finite cutoff,

$$\begin{aligned} \partial_i[\Theta_a(U)\partial_i U^a] &= \partial_b \Theta_a(U)(\partial_i U^b \partial_i U^a - \Gamma_i^{ba}) \\ &\quad + \Theta_a(U) \diamond \partial_i^2 U^a, \end{aligned} \quad (4.12)$$

where

$$\Theta_a(U) \diamond \partial_i^2 U^a := \Theta_a(U) \partial_i^2 U^a + \Gamma_i^{ba} \partial_b \Theta_a(U). \quad (4.13)$$

The two covariance terms in (4.12) cancel exactly. Keeping only the first line or citing a balanced derivative-square jet while deleting (4.13) would break the finite-cutoff identity.

5 Complete linear payment

Apply the theorem entrywise to (2.7). For every finite p and $\delta > 0$,

$$\mathcal{J}_r^{\text{lin}} \in L^p(\Omega; H^{-1/2-\delta}), \quad \mathbb{Q}_r^{\text{lin}} \in L^p(\Omega; H^{-3/2-\delta}). \quad (5.1)$$

The R-068 deterministic payment extends without change of architecture. For $1 < s < 2$, its low and high endpoints give

$$\|\Pi(A)\|_{H^s} \leq C X^{s/2} Y^{(2-s)/6}, \quad (5.2)$$

where

$$X = \sum_j \|A_j\|_{H^2}^2, \quad Y = \left\| \left(\sum_j |A_j|^2 \right)^{1/2} \right\|_6^6.$$

If $M_s = \|Q\|_{H^{-s}}$, three-factor weighted Young gives

$$\boxed{|\langle Q, \Pi(A) \rangle| \leq \eta X + \zeta Y + C_s \eta^{-\frac{3s}{2(2-s)}} \zeta^{-1/2} M_s^{3/(2-s)}}. \quad (5.3)$$

Set $s = 3/2 + \delta$, $0 < \delta < 1/2$. Then (5.1)–(5.3) pay the entire symmetric linear frame, including the polynomial and rational, pp and non- pp pieces, in one global allocation. At $\delta = 1/10$,

$$|\mathcal{L}_j^{\text{sym}}| \leq \eta X + \zeta Y + C \eta^{-6} \zeta^{-1/2} \|\mathcal{J}^{\text{lin}}\|_{H^{-3/5}}^{15/2}. \quad (5.4)$$

Here the displayed norm is that of the undifferentiated current; its divergence is tested in $H^{-8/5}$. Every required moment is supplied by (4.3).

The R-070 Cartan coefficient is also a finite smooth matrix one-form in U . The same theorem gives a model $\mathbb{J}^{\text{Car}}$ in $H^{-1/2-\delta}$. The deterministic interpolation used in R-070, now at $\kappa = 1/2 + \delta$, yields

$$\|A \wedge DA\|_{H^{1/2+\delta}} \leq C X^{(5+2\delta)/6} Y^{(1-2\delta)/18}. \quad (5.5)$$

Therefore

$$\boxed{|\langle \mathbb{J}^{\text{Car}}, A \wedge DA \rangle| \leq \eta X + \zeta Y + C \eta^{-\frac{3(5+2\delta)}{2(1-2\delta)}} \zeta^{-1/2} \|\mathbb{J}^{\text{Car}}\|_{H^{-1/2-\delta}}^{9/(1-2\delta)}}. \quad (5.6)$$

At $\delta = 1/10$, the negative η exponent is $39/4$ and the model moment is $45/4$. These are finite. Equations (5.4) and (5.6) replace the false $\kappa = 1/10$ attributions in R-070 and close its full weighted linear subgate.

6 The nonlinear terminal square has kernel leakage

For one generator and direction suppress indices and put

$$z = U, \quad a = A, \quad y = G_i, \quad b = C_i,$$

$$M_0 = M(z), \quad M_1 = M(z + a), \quad E = M_1 - M_0 - DM(z)[a], \quad w = M_0^T y, \quad (6.1)$$

$$\delta = M_1^T(y + b) - M_0^T y. \quad (6.2)$$

Combining the $G \otimes G$ part of the coefficient remainder with $\mathcal{N}_J$, the cross, and the terminal square gives the exact local identity

$$\sum_r \left\{ \langle w_r, Q_{\text{II}} E_r^T(y + b) \rangle + \frac{1}{2} |\delta_r|_{Q_{\text{II}}}^2 \right\}. \quad (6.3)$$

The remaining covariance part is

$$-\frac{1}{2} [B(z + a) - B(z) - DB(z)[a]] : \Gamma_i, \quad (6.4)$$

which is independent of b .

The following six-real production fixture is exact:

$$\begin{aligned} z &= (-1, -1, 0, 0, -1, 0), \\ x = z + a &= (1, -1, 0, 1, 0, 0), \\ a &= (2, 0, 0, 1, 1, 0), \quad y = (1, 0, 0, 0, 0, 0), \\ n &= (-1, 0, 0, 1, -1, 0), \quad b = tn. \end{aligned} \quad (6.5)$$

For every $e > 0$ and every $r = 1, 2, 3$,

$$M_r(x)^T n = 0. \quad (6.6)$$

Indeed $n \cdot x = n \cdot S_r x = 0$. Hence the terminal current square in (6.3) is independent of t , and its $c = M_1^T b$ cross and square pieces vanish. Direct production-frame algebra gives

$$n \cdot \sum_r E_r Q_{\text{II}} w_r = \frac{27(6e^2 + 22e + 27)}{400(e + 3)^3} > 0. \quad (6.7)$$

At $e = 10^{-12}$ this coefficient is 0.06749999999997061. Consequently

$$\mathcal{R}_{\text{shift}}(z, a, y, tn) = C(z, a, y, \Gamma) + t, c_e, \quad c_e > 0, \quad (6.8)$$

and it is unbounded below as $t \rightarrow -\infty$.

This is not a counterexample to an integrated estimate: a spatial field with large $b = DA$ also spends Cameron–Martin energy. It does prove that the terminal square alone cannot absorb $\mathcal{N}_J$, and that no pointwise or derivative-energy-free version of the R-070 successor can hold. Even adding back the separated linear frame does not remove the need for an energy or expectation mechanism; the old-frame first variation carries the compensating derivative displacement in the exact untranslated identity.

7 Evidence map and refined successor

The proof route is now:

1. **Closed:** R-050 supplies the common geometric second level. The L^2 Littlewood–Paley version of the same reconstruction proves the matrix one-form lift (4.3).

2. **Retired:** literal raw $\mathbb{Q}_r^{\text{lin}} \in H^{-1-1/10}$, including the nominally paid pure pp attribution. The low-high term (3.2) is nonzero.
3. **Closed:** extend R-068 from $s = 1 + \kappa$ to every $1 < s < 2$. At $s = 8/5$ the full symmetric frame is paid with moment 15/2.
4. **Corrected and closed:** the Cartan current is paid at $H^{-3/5}$, with moment 45/4, rather than at $H^{-1/10}$ with moment five.
5. **Retired:** pointwise absorption of the nonlinear coefficient remainder by the terminal square. Fixture (6.5)–(6.8) exposes an exact kernel leakage.
6. **Still open:** an integrated strict-past kernel-leakage estimate. It must use the Cameron–Martin H^2 cost and causal frequency structure, while retaining the range-visible part inside the positive square and the bad/tip region as a whole secant. It may not invoke Malliavin derivatives of the control or count one random constant per shell.

The smallest honest next statement is therefore a range/kernel grouped form: decompose the nonlinear derivative-displacement coefficient relative to the terminal stacked frame. Keep its range component with $|M(U+A)^T(G+DA) - M(U)^T G|_{Q_H}^2/2$. Pay the kernel component only after spatial integration, with one arbitrary Cameron–Martin allocation, one terminal sextic allocation, and one integrable random constant. The decomposition must be implemented without a singular inverse at the cone tip; the R-068 good/bad whole-secant architecture remains load-bearing.

8 Devil’s-advocate review

1. **UPHELD against the old target:** “The non- pp current is a balanced $H^{-1-1/10}$ model.” The coefficient (2.6) and low-high block (3.2) give the divergent lower bound (3.3).
2. **UPHELD against the old pure- pp payment:** “Covariance centering turns $\Delta(U^T S_r U)$ into an $H^{-1-1/10}$ model.” Centering does not remove its nonzero-frequency low-high principal symbol. The term is paid at $H^{-3/2-\delta}$ instead.
3. **DISMISSED:** “The weaker regularity loses one-use.” R-068 already permits every $s < 2$. Equation (5.3) requires moment 15/2 at $s = 8/5$, and the one-form lift supplies every finite moment.
4. **VALID with mitigation:** “A6 only stated one special current in $\mathcal{C}^{-1/2-}$.” Section 4 records the needed matrix-valued $H^{-1/2-}$ strengthening. It uses exactly the same canonical second level, but the L^2 estimate is written out rather than inferred from the Holder statement.
5. **DISMISSED:** “A new current counterterm is hidden in the divergence.” Evenness gives (4.11), and the two covariance terms in (4.12)–(4.13) cancel exactly.
6. **UPHELD against the R-070 Cartan exponent:** “The Cartan model is $H^{-1/10}$ with a fifth moment.” Its generic low-high regularity is $H^{-1/2-\delta}$. Equation (5.6) repairs the payment with finite moment 45/4 at $\delta = 1/10$.
7. **UPHELD against pointwise Schur closure:** “The terminal square controls every derivative displacement.” The vector n in (6.5) is in the common terminal-frame kernel, while (6.7) is strictly positive.
8. **VALID with mitigation:** “The kernel fixture disproves the desired stochastic theorem.” It disproves only a pointwise or derivative-energy-free bound. Integrated H^2 absorption and strict-past frequency gains remain available and form the next gate.

9. **UPHELD against termwise absolute payment:** “The leakage can be bounded as a generic $\mathbb{J}A^2DA$ term.” The scaling has the critical allocation $X^{3/4}Y^{1/4}$ at the endpoint and leaves no exponent for an unbounded random norm. The range square and causal grouping must remain.
10. **UPHELD against closure:** “The full linear payment proves controlled-shell one-use.” The nonlinear integrated kernel leakage, finite-energy extension, umbrella one-use estimate, and Nelson theorem remain open. No tier promotion follows.

Result footer

The primary executable derives the weighted frame split and Sobolev/Young exponents from the production source, evaluates the low–high obstruction, and checks the exact kernel fixture. The non-importing executable rebuilds the frames and the same diagnostics independently. The integrated wrapper hash-pins the A6, R-068, R-070, source, note, and PDF authorities before child execution.

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Result ID: A13-CLASSII-ONE-FORM-SOBOLEV-LINEAR-CLOSURE-AND-KERNEL-LEAKAGE
Precise statement:      At fixed positive floor, the R-050 enhancement
                        reconstructs the full matrix one-form in
                         $H^{-1/2-\delta}$ ; generalized symmetric and Cartan
                        interpolation pays the complete linear frame.
Scope:                  Smooth finite matrix one-forms; common real-even
                        spectral approximation; regular strict-past class.
Dependencies:           A1 production functional; R-050; R-068; R-070.
Evidence grade:          ANALYTIC, EXACT, EXECUTED.
Reproduction command:   python codes/foundations/
                        a13_classii_one_form_sobolev_linear_closure_verify.py
Expected output:         primary 21/21; independent 21/21; integrated 35/35;
                        aggregate 77/77; exit 0.
Falsification gate:      A low-high symbol, one-form lift, interpolation,
                        kernel/slope, hash, PDF QA, or assertion fails.
Tier before / after:     T4 / T4; no promotion.
No-overclaim statement:  Pointwise square closure is retired, but integrated
                        strict-past leakage, finite-energy, one-use, Nelson,
                        and every removal limit remain open.
Next required action:    Prove the integrated strict-past Cameron--Martin/
                        frequency bound for terminal-frame kernel leakage.

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This result neither constructs an interacting measure nor removes the density floor, regulator, or volume cutoff. It proves no phase transition, BCC selection, T5, T6, or T7 statement.